

We built a reconciliation data platform — then used it to test how AI should review code

A real client needed reconciliation data made legible. We shipped a production platform for the **Indigenomics Institute** — and its own **30 merged pull requests** became the industrial dataset for a pre-registered study of multi-agent AI code review.

Research companion: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

En-Ping Su · Tong Wu · Shiting Huang · Mengshan Li

Khoury College of Computer Sciences, Northeastern University · Vancouver · Advisor: Yvonne Coady · Industry Partner: Indigenomics Institute (Carol Anne Hilton, Founder)

RESEARCH CONTRIBUTION

More AI reviewers don't help — **decorrelation does**. Agreement predicts truth only when the reviewers are independent.

89%

3 independent families agree → matches ground truth

54%

one model, three runs → mostly noise

01 THE PROJECT — RAP DATA PORTAL

PROBLEM

Companies publish reconciliation commitments. Legal precedents sit scattered across court records. There is no searchable, verifiable, unified view — so nobody can track progress.



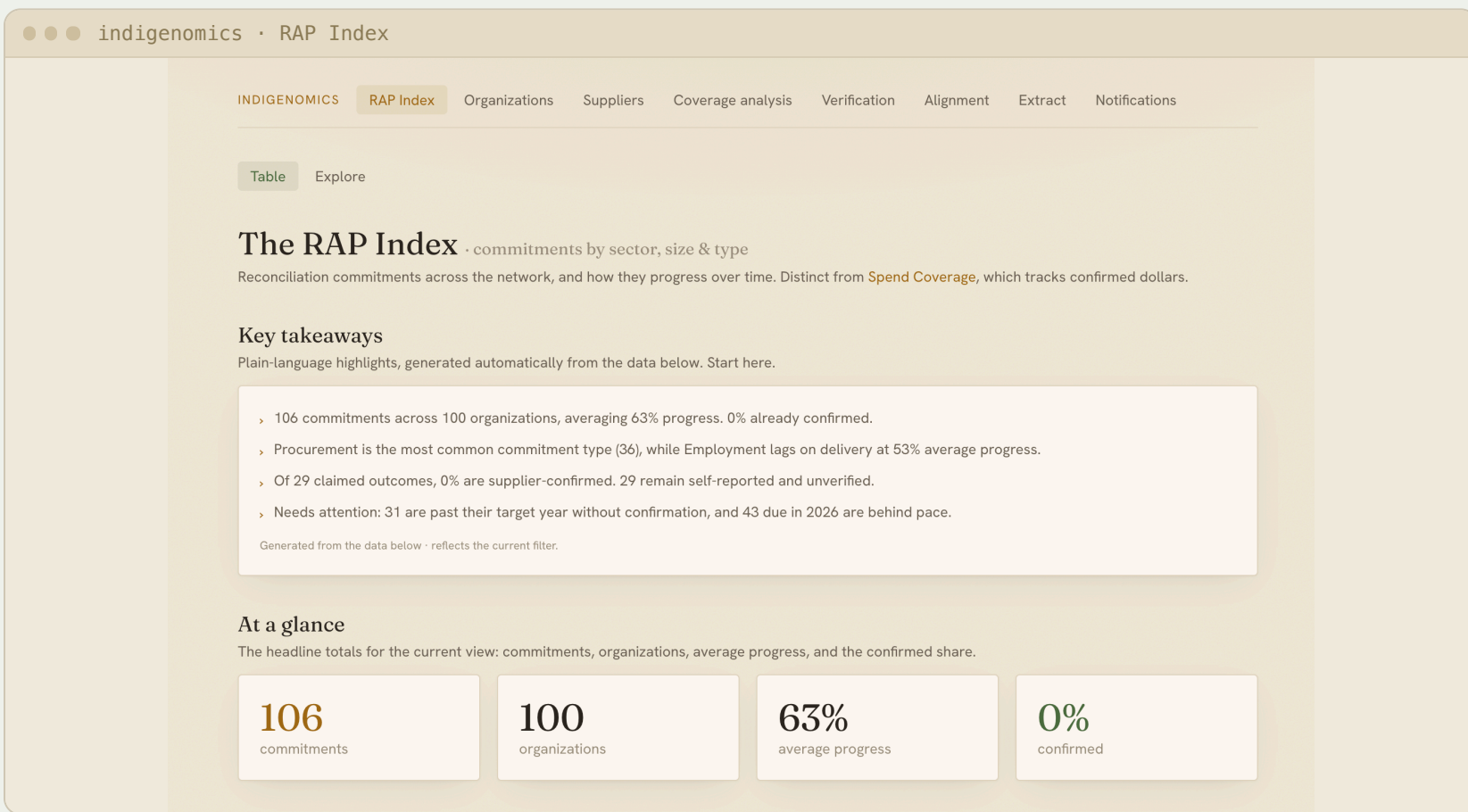
OUR SOLUTION

- Searchable RAP commitments
- Legal precedent database
- AI extraction from PDFs
- Human review workflow
- AWS serverless hosting
- Authentication & roles
- Progress tracking
- Coverage analytics



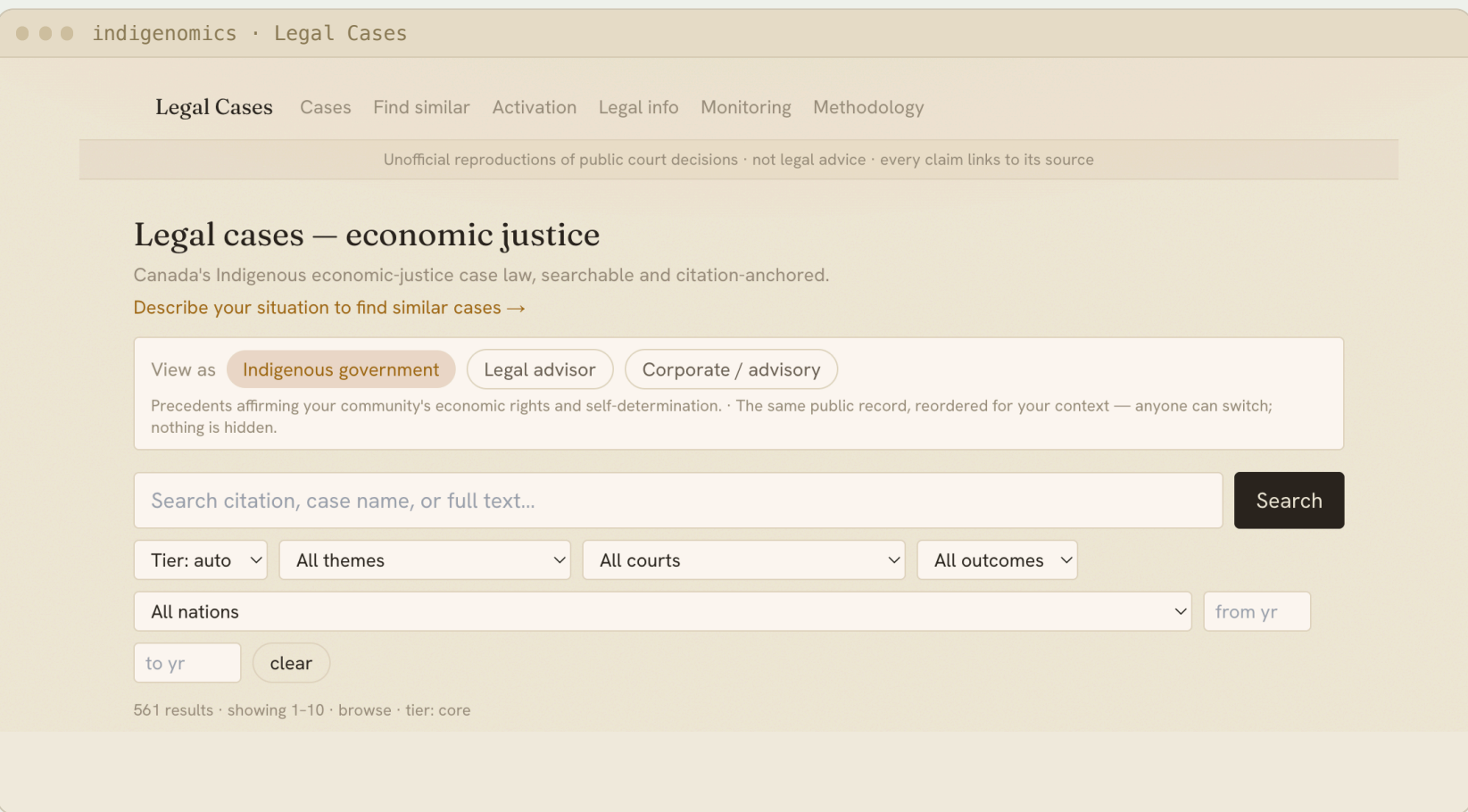
WHO USES IT

- Indigenomics Institute
- Indigenous organizations
- Corporate partners
- Researchers & legal advisors



RAP Index

- 106 commitments · 100 organizations
- 63% average progress tracked



Legal Cases

- 561 economic-justice decisions
- citation-anchored, role-based views

02 HOW IT BECAME RESEARCH

Client problem

reconciliation data scattered & unverifiable



We built the portal

production platform, shipped on AWS



179 pull requests

30 of them ours — real, merged, AI-reviewed



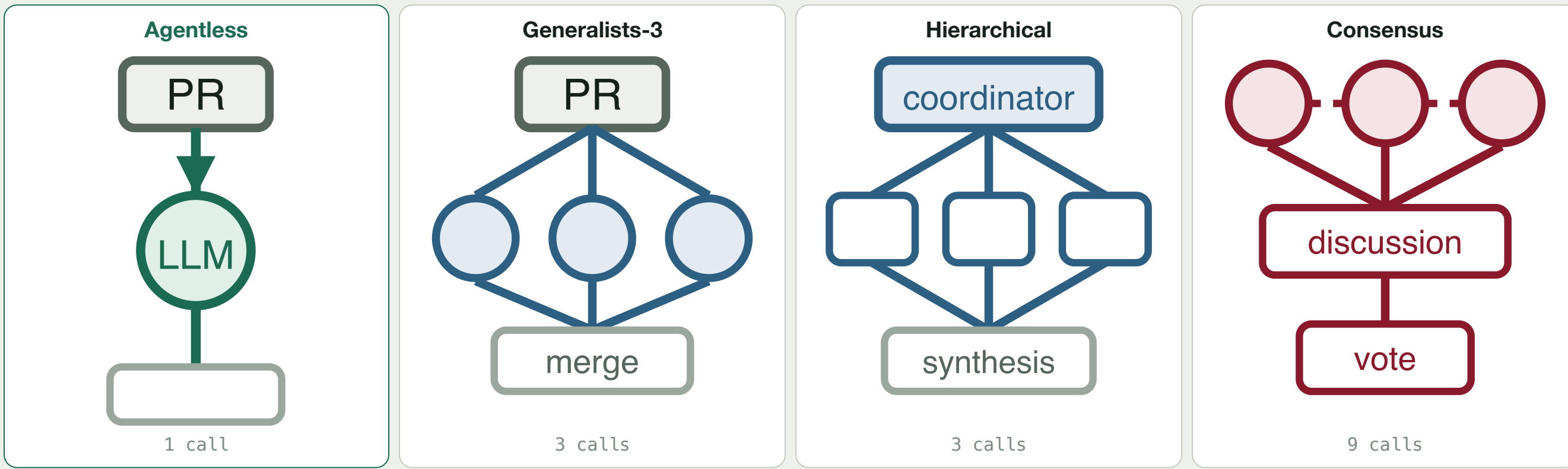
Benchmark dataset

E1 Qodo · E2 SWE-PRBench · E3 our portal — 2,415 reviews



Four AI review architectures

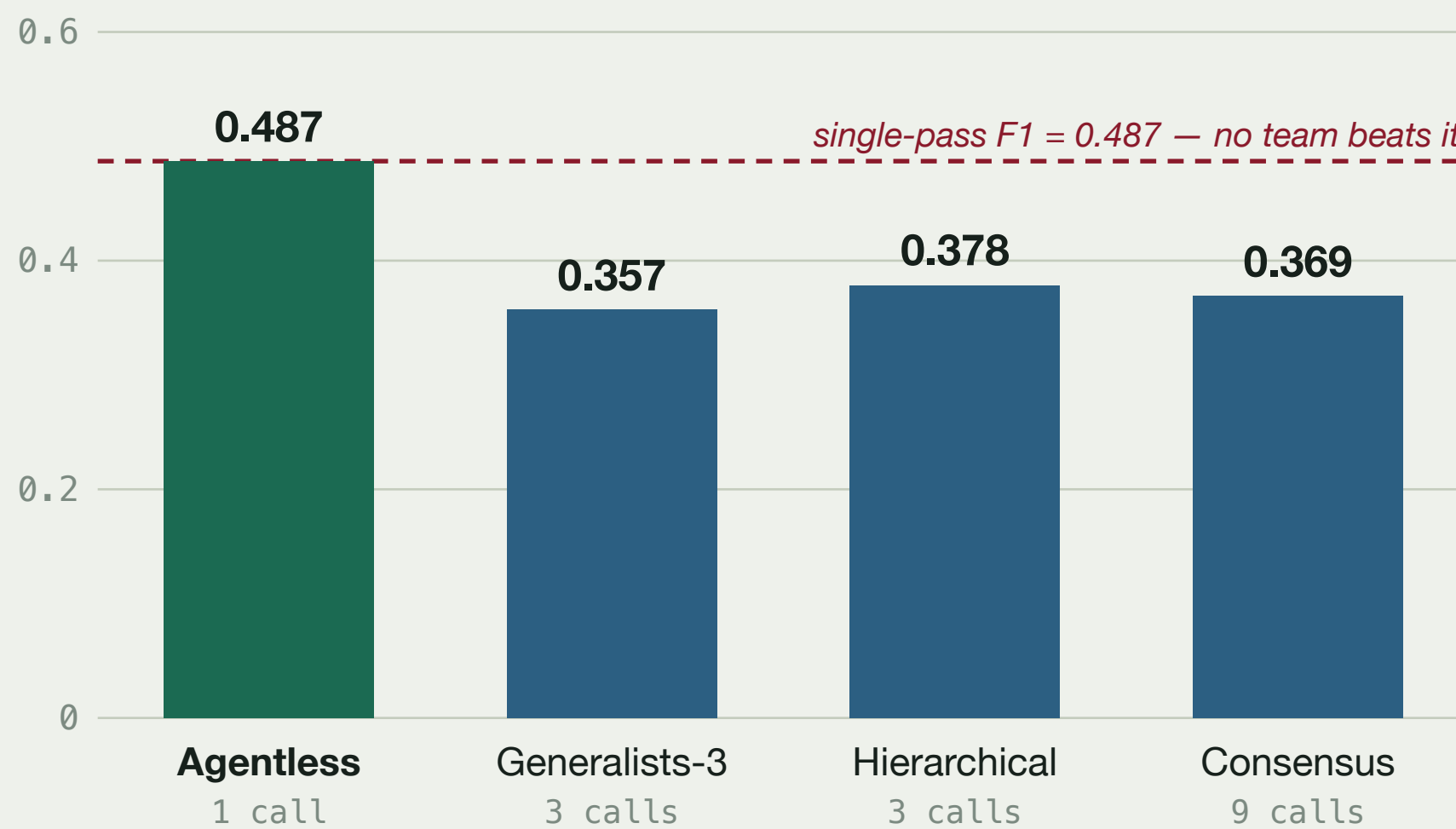
each rung adds exactly one capability



Held constant: same PR snapshot · same model (Claude Haiku 4.5) · same prompts · three runs per arm · pre-registered on OSF.

03 WHAT WE FOUND · E1 · Q000, 99 PRs

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling ($p=0.73$); peer debate adds nothing over a manager ($p=0.12$).



Extra agents buy recall at a price — up to **5x** the cost per confirmed finding.

Agentless
1 call

0.34

Generalists-3
3 calls

0.45

Hierarchical
3 calls

0.50

Consensus
9 calls

1.76

5x more expensive per confirmed finding than single-pass review

Bounded. Recall plateaus at ≈ 0.83 ; on our own unlabeled code, LLM judges disagree ($\kappa=0.03$).

PROJECT IMPACT

- Production platform **delivered** to the Indigenomics Institute
- Reconciliation commitments made **searchable and verifiable**
- Indigenous legal precedent **explorer**, citation-anchored

RESEARCH IMPACT

- First **compute-matched** comparison of review architectures
- **Independence beats more agents** — decorrelate your verifiers
- Guidance for anyone deploying AI code review in production



Code & data
rap-review-research



Live portal
try the demo